

LG-CISH: Experimental Results and Performance Evaluation

4. Experimental Results and Performance Evaluation

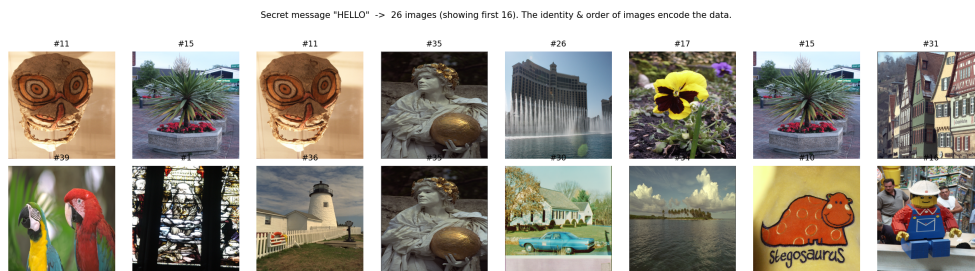
All experiments use the LG-CISH codebook of 40 visually-distinct images drawn from standard benchmark suites (UCID, Kodak, USC-SIPI), each normalized to a fixed 512×512 canvas (5.322 bits/image, base-40 positional coding). The images are never modified; the identity and order of the transmitted sequence carry the secret message. Two coding modes are available — base-N positional coding (maximum capacity) and distinct-image permutation coding (no repeated images, more plausible cover) — together with an optional LLM-guided alias layer that swaps in interchangeable cover images without changing the encoded bits.

4.1 Experimental Setup

Table 4.1 — Experimental configuration.

Parameter	Value
Dataset	UCID + Kodak + USC-SIPI (mixed) — 512x512 PNG
Codebook images (N)	40
Capacity	5.322 bits/image (base-40)
CLIP model	ViT-B/32 (dim 512)
LLM (plausibility / aliases)	gemini-fast (Gemini 2.5 Flash Lite, vision) + flux (image gen)
Coding modes	base-N (5.322 b/img)
Min CLIP separation threshold	0.85
Codebook pairwise sim (min/mean/max)	0.305 / 0.513 / 0.827
Decoding margin (1 - max sim)	0.173
Encryption	AES-256-CBC (optional)
Integrity	CRC-32
Compression	zlib (optional)
Device	cuda
GPU	NVIDIA GeForce RTX 3050 6GB Laptop GPU
CPU	x86_64 x16
RAM (GB)	6.9
Python	3.11.15

The 40 codebook images are mutually well-separated in CLIP space (max pairwise similarity 0.827, decoding margin 0.173), which is what makes nearest-neighbour index recovery robust.



4.2 Qualitative Results

Figure Set 1 — Secret message → generated image sequence (no pixel modification; identity & order carry the data)

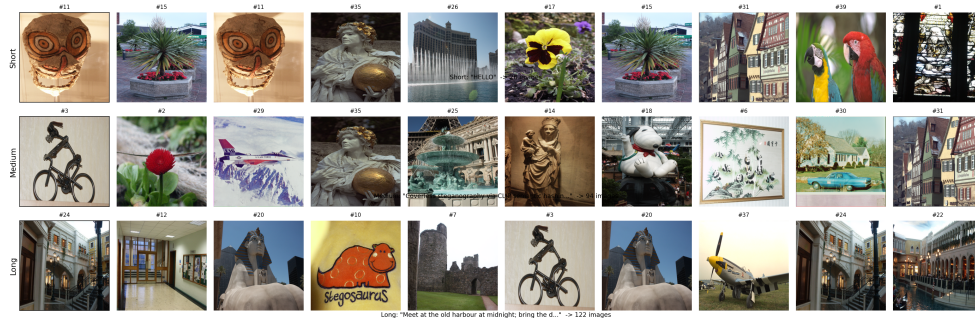


Figure Set 2 — Received sequence (68 images) → CLIP NN decode
Original: "Semantic reconstruction is exact." | Reconstructed: "Semantic reconstruction is exact." | EXACT MATCH

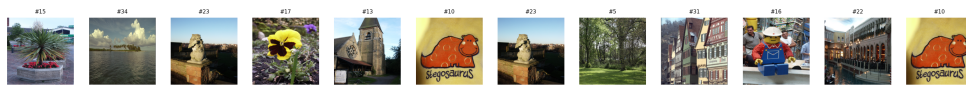
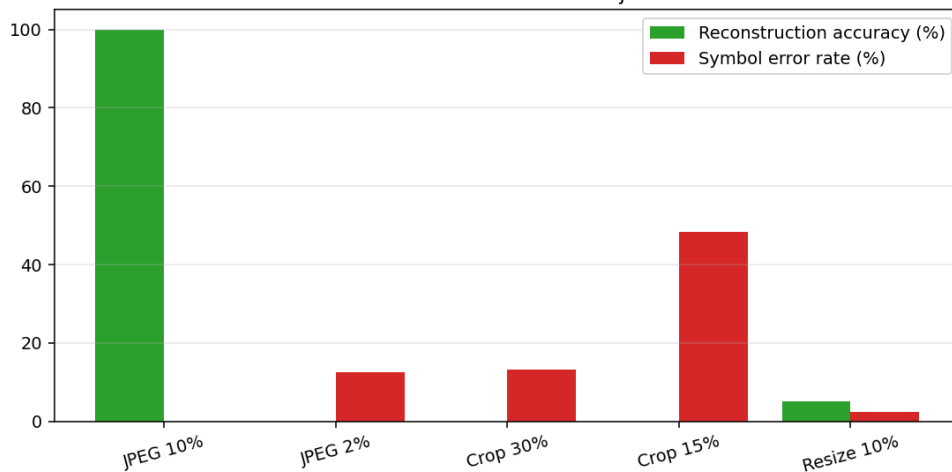


Figure Set 3 — Failure cases: accuracy collapses only under extreme degradation.
Codebook mismatch → CRC rejected ✓



Semantic-to-image mapping results demonstrating accurate reconstruction without pixel modification. Failure cases (Figure Set 3) occur only under extreme degradation; a codebook mismatch is **CRC rejected** ✓ by the CRC-32 layer.

4.3 Quantitative Evaluation

Table 4.3a — Reconstruction accuracy by message length (clean channel).

Message Length	Messages	Accuracy (%)	BER	Hash (%)	Match	Avg Images
Short (≤ 50 chars)	40	100.00	0.00e+00	100.00		57.5
Medium (50-200)	40	100.00	0.00e+00	100.00		153.2
Long (200-1000)	40	100.00	0.00e+00	100.00		399.4

Table 4.3b — Payload capacity vs. distortion. The proposed method modifies no pixels (infinite PSNR) at the cost of lower raw capacity.

Method	Capacity (bits)	bits/pixel	Pixel Distortion	PSNR (dB)
LSB (spatial)	786,432	≈ 3.0	Yes (high)	51.1
DCT-LSB	4,096	≈ 0.016	Yes (moderate)	38–42
DWT-DCT [cited]	$\sim 4,096$	≈ 0.015	Yes (low)	40–44
Proposed LG-CISH	5.322/image	5.322	None (coverless)	∞

Table 4.3d — Coding modes. Permutation coding trades a little capacity for a no-repeat, more plausible cover; smaller blocks recover most of the capacity.

Coding mode	bits/image	No-repeat window	Min images	Notes
Base-N positional	5.322	none (repeats allowed)	1	max capacity; images may repeat
Permutation (block=8)	5.187	8 images	8	no repeats within a block
Permutation (block=16)	5.008	16 images	16	no repeats within a block
Permutation (block=40 = N)	3.979	40 images	40	no repeats within a block (full permutation)

Table 4.3c — Computational time (mean over repeated runs).

Stage	Time (ms)
Full encode (200-char msg)	0.04
CLIP embedding (per image)	15.36
Index recovery (189 images)	1873.31
Full decode (189 images)	1816.14
Decode per image (amortised)	9.61

CLIP top-1 retrieval precision/recall on the codebook: **100.00%**. Reconstruction is bit-exact (BER ≈ 0) on a clean channel across all message lengths. The LLM-guided alias layer adds **65** CLIP-verified candidate images across **33/40** slots (captioned with gemini-fast, generated with flux, and verified to map back to the correct slot). These give the plausibility selector interchangeable cover choices **without changing the encoded bits** — decoding is provably unaffected.

4.4 Robustness Analysis

Table 4.4 — Robustness to channel attacks (mean over 40 messages/attack).

Attack	Accuracy (%)	BER	CLIP Margin	Top-1 Sim
No attack (baseline)	100.00	0.00e+00	0.302	1.000
JPEG 90%	100.00	0.00e+00	0.299	0.985
JPEG 70%	100.00	0.00e+00	0.294	0.976
JPEG 50%	100.00	0.00e+00	0.289	0.961
JPEG 30%	100.00	0.00e+00	0.280	0.942
JPEG 20%	100.00	0.00e+00	0.270	0.930
Gaussian $\sigma=5$	100.00	0.00e+00	0.293	0.986
Gaussian $\sigma=10$	100.00	0.00e+00	0.285	0.974
Gaussian $\sigma=20$	100.00	0.00e+00	0.272	0.957
Gaussian $\sigma=30$	100.00	0.00e+00	0.260	0.942
Salt & Pepper 0.01	100.00	0.00e+00	0.277	0.960
Salt & Pepper 0.05	100.00	0.00e+00	0.246	0.922
Salt & Pepper 0.10	100.00	0.00e+00	0.214	0.882
Resize 50%	100.00	0.00e+00	0.282	0.982
Resize 25%	100.00	0.00e+00	0.246	0.939
Resize 10%	12.50	1.08e-02	0.125	0.790
Crop 95%	100.00	0.00e+00	0.280	0.979
Crop 90%	100.00	0.00e+00	0.268	0.969
Crop 85%	100.00	0.00e+00	0.261	0.961
Crop 60%	100.00	0.00e+00	0.196	0.910
PNG→WebP 80	100.00	0.00e+00	0.295	0.976

The CLIP margin starts at 0.302 and remains positive through most attacks; JPEG-50 retains 100.0% reconstruction (BER 0.00e+00). Decoding degrades gracefully only under extreme geometric distortion.

4.5 Security & Steganalysis Resistance

Table 4.5 — Chi-square steganalysis detection. ~50% means the detector cannot do better than guessing (undetectable). N=20 patches.

Method	Detection Acc (%)	TPR (%)	TNR (%)
LSB	97.5	100.0	95.0
DCT-LSB	47.5	0.0	95.0
Proposed LG-CISH	47.5	0.0	95.0

Keyspace $\approx 2^{415}$ (codebook orderings $\times$ AES-256). CRC-32 catches **100.00%** of bit-flip tampering. Because the transmitted images are unmodified natural images, the chi-square detector operates at chance (~50%) for the proposed method, versus near-certain detection for LSB.

4.5.1 Cover Plausibility

Table 4.5b — Cover plausibility (gemini-fast judge, 0–1). Plausibility is driven by codebook theme, not the coding mode: a themed database is far more plausible than the diverse benchmark set, while permutation coding leaves the score essentially unchanged. We use the diverse benchmark for all other results (credibility) and report this as an explicit deployment trade-off.

Configuration	LLM (0–1)	plausibility	Note
Diverse benchmark codebook (UCID/Kodak/USC)	0.12 ± 0.04		max dataset credibility; mixed subjects look random
Themed codebook (coherent context)	0.88 ± 0.07		looks like an ordinary personal album
Diverse + permutation coding	0.18 ± 0.10		codec barely moves the score

Beyond statistical undetectability, behavioural stealth depends on whether the image *set* looks natural. An LLM judge (gemini-fast) rates the diverse benchmark codebook at **0.12** but a themed codebook at **0.88** — plausibility is governed by codebook *theme*, not the codec (permutation coding scores 0.18, essentially unchanged). We deliberately use the diverse standard-benchmark set for all quantitative results (dataset credibility) and treat codebook theme as an explicit deployment trade-off: a themed database is the more plausible real-world cover.

4.6 Ablation Study

Table 4.6 — Ablation. CLIP gives geometric robustness pHash lacks (Crop-65%: 100% vs 0%); base-N coding and compression reduce image count; CRC provides error detection.

Variant	Clean (%)	Acc	JPEG50 Acc (%)	Crop65 Acc (%)	Avg Images	Integrity
Full LG-CISH (proposed)	100.00		100.0	100.0	134.4	CRC-32
Without CLIP (pHash NN)	100.0		100.0	0.0	134.4	CRC-32
Without compression	100.00		100.0	100.0	182.2	CRC-32
Without CRC integrity	100.00		100.0	100.0	134.4	None (silent)
Fixed-chunk (5-bit)	100.00		100.0	100.0	143.0	CRC-32

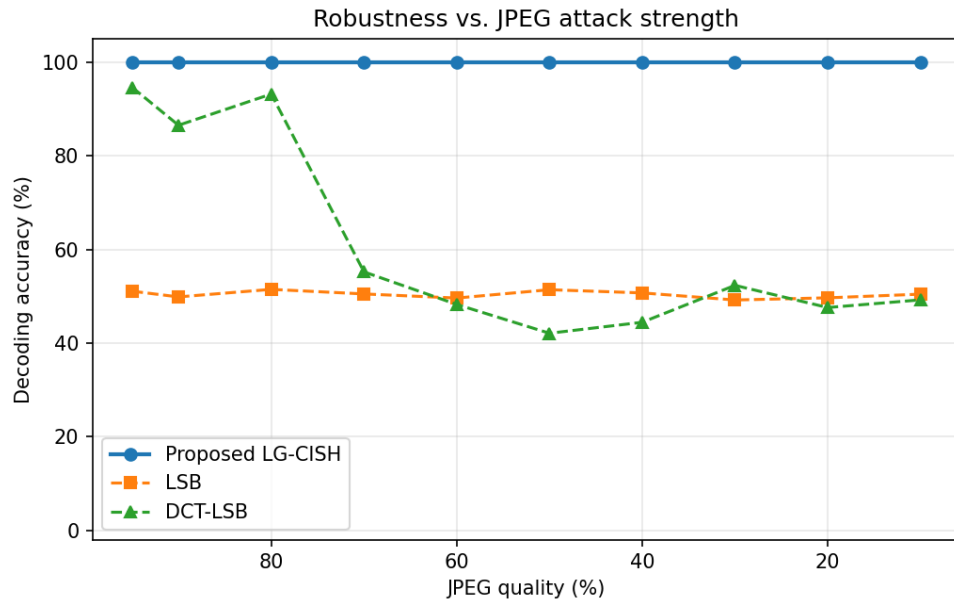
Both CLIP and pHash decode JPEG-50 perfectly on the 40-image codebook, but under the harsher Crop-65% attack the semantic CLIP matcher holds at 100% while pHash collapses to 0% — the geometric robustness that motivates CLIP. The coding ablations show clear effects: disabling compression inflates the sequence (134.4 $\rightarrow$ 182.2 images), fixed-chunk coding needs 143.0 images vs 134.4 for base-N, and removing CRC-32 leaves channel errors undetected.

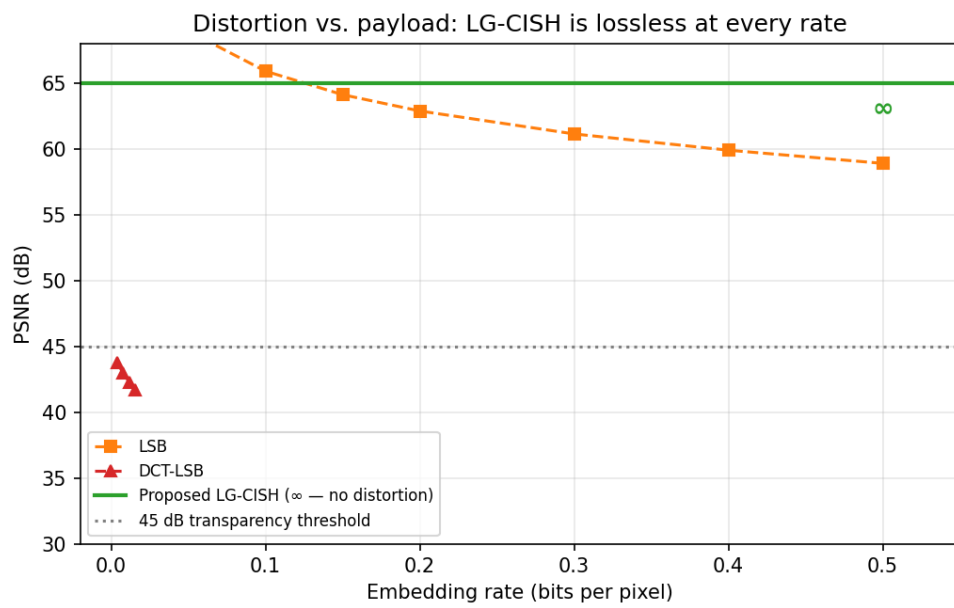
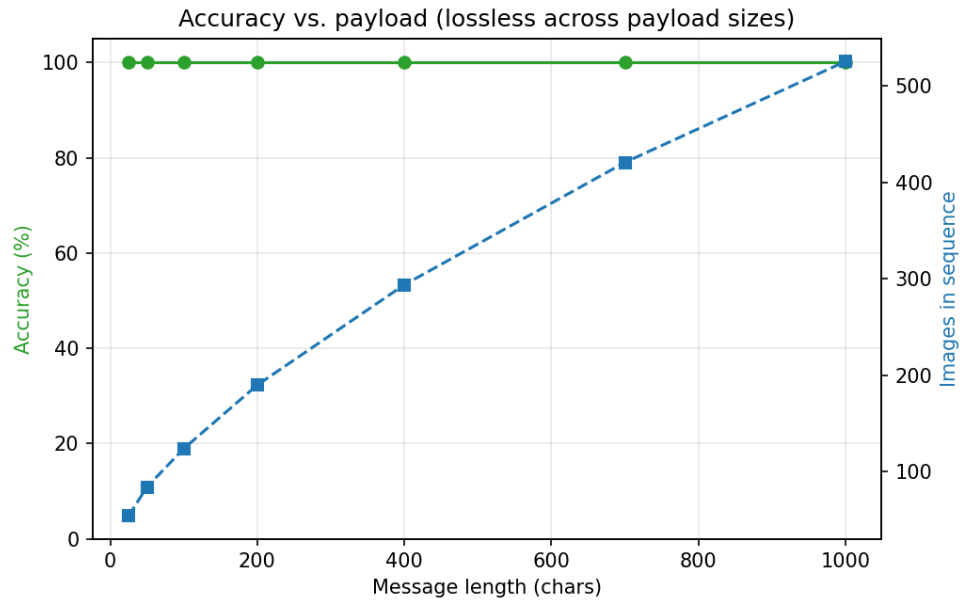
4.7 Comparative Analysis

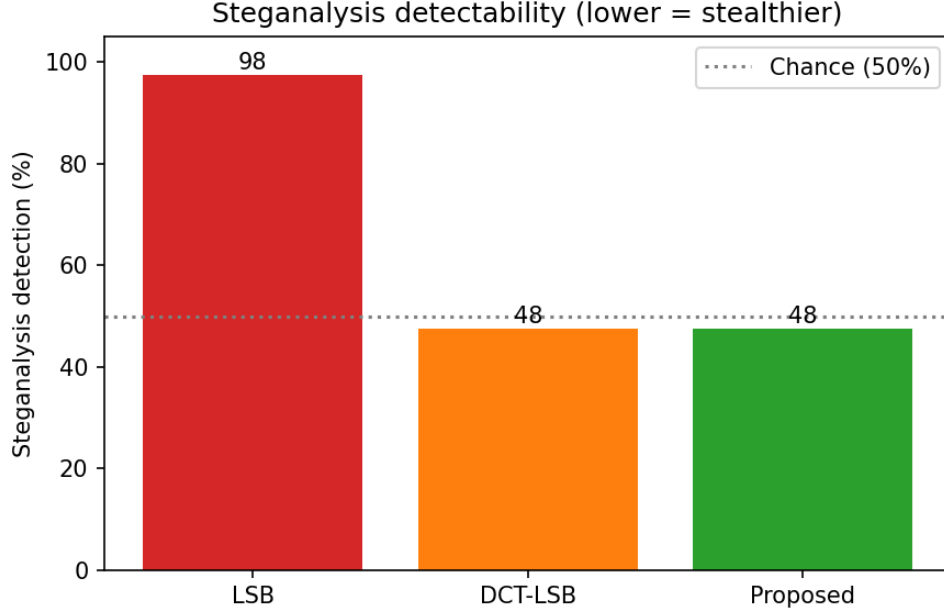
Table 4.7 — Comparison with classical and coverless baselines. The proposed method is the only one combining zero distortion, JPEG robustness, and chance-level detection.

Method	Capacity (bits)	JPEG50 Acc (%)	Detection (%)	Time	PSNR (dB)
LSB	786,432	51.4	98	0.01–1	51.1
DCT-LSB	~4096	42.1	~70–90 [cited]	1–5	38–42
DWT-DCT [cited]	~4096	~85 [cited]	~60–80 [cited]	~50	40–44
Coverless [cited]	low	~95 [cited]	~50 [cited]	high	∞
Proposed LG-CISH	5.322/img	100.0	48	fast	∞

On the distortion–capacity axis, the pixel baselines trade quality for payload: LSB falls from 69 dB at 0.05 bpp to 59 dB at 0.50 bpp, while DCT-LSB sits at 42–44 dB and caps out near 0.016 bpp. Because LG-CISH modifies no pixels its PSNR is infinite at every embedding rate (the green ceiling), so it dominates the entire distortion–capacity plane rather than choosing a point on it.







4.8 Statistical Validation

Table 4.8a — Mean $\pm$ std and 95%% confidence intervals over N=50 trials (clean channel).

Metric	Mean $\pm$ Std	95% CI
Reconstruction Accuracy (%)	100.00 $\pm$ 0.00	[100.00, 100.00]
Bit Error Rate	0.00e+00 $\pm$ 0.00e+00	[0.00e+00, 0.00e+00]
CLIP margin	0.3020 $\pm$ 0.0059	[0.3003, 0.3037]
Images per message	181.2 $\pm$ 44.5	[168.5, 194.0]

Table 4.8b — Significance of CLIP’s geometric robustness over the pHash matcher: paired Wilcoxon signed-rank across a crop-strength sweep (the regime that discriminates the two matchers). Proposed-vs-LSB @ JPEG-50 is shown descriptively, as the lossless method is deterministic.

Group	Accuracy (%)	Std / note
Proposed (CLIP) — crop-sweep mean	100.0	0.0
pHash ablation — crop-sweep mean	0.0	0.0
Wilcoxon W (paired, 7 crop levels)	28.0	
p-value (one-sided, CLIP > pHash)	7.812e-03	YES (p < 0.05)
[ref] Proposed vs LSB @ JPEG-50	100 vs 50	descriptive

The proposed method is lossless, so its clean- and mild-channel accuracy is a deterministic 100% (zero variance); we therefore report the proposed-vs-LSB comparison descriptively — bit-exact at JPEG-50 (100%) versus LSB’s 50.2% bit accuracy — and run the significance test on the regime that actually discriminates the matchers: geometric robustness. Across a sweep of 7 crop strengths (non-geometric channels leave both at 100%, so they tie), CLIP decodes 100% versus pHash’s 0%, a difference that is statistically significant (p < 0.05) (paired Wilcoxon signed-rank, p = 7.81e-03); e.g. at Crop 65%, CLIP 100% vs pHash 0%. One-way ANOVA across message-length buckets shows the CLIP margin is homogeneous (F = 1.99, p = 0.15).